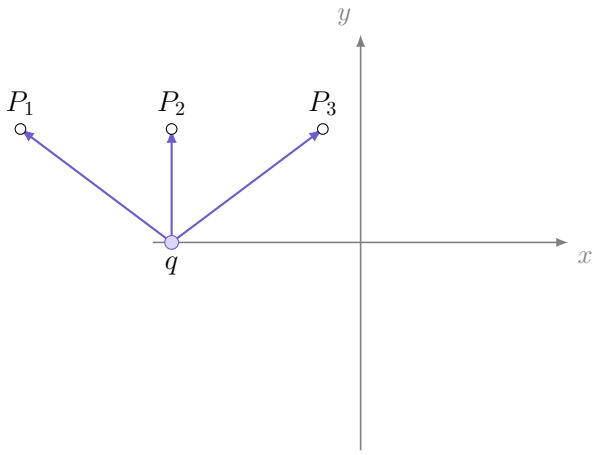
Intro to Biot–Savart (completed)

Goal.

- Warm up with something you already know: superpose the $\vec{E}$ field of a point charge at three locations, being explicit about separation vectors and unit vectors.
- Reuse that exact same geometry to introduce the Biot–Savart law for a single *moving* point charge, $\vec{B}$.
- Compare $\vec{E}$ and $\vec{B}$ at those same three points — same source, same locations, very different field.
- Scale up to three moving charges in a line and see their $\vec{B}$ contributions add (not cancel) at an off-axis point.
- Make the leap from one moving charge ($q\vec{v}$) to a continuous current ($I\,d\vec{\ell}$) — the form of Biot–Savart we'll actually use for wires, loops and solenoids.

1. Warm-up: $\vec{E}$ from a point charge

A charge $q = +2.0\ \mu\text{C}$ sits at $(-5, 0, 0)$ cm. Find $\vec{E}$ at three field points: $P_1 = (-9, 3, 0)$, $P_2 = (-5, 3, 0)$, $P_3 = (-1, 3, 0)$ (all in cm).



Sketch the three separation vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3$ from q to each field point on the diagram above.

Step 1 — separation vectors $\vec{r}_i = \vec{r}_{P_i} - \vec{r}_q$ (cm):

$\vec{r}_1 = (-4,\ 3,\ 0)$

$\vec{r}_2 = (0,\ 3,\ 0)$

$\vec{r}_3 = (4,\ 3,\ 0)$

Step 2 — magnitudes and unit vectors $\hat{r}_i = \vec{r}_i / r_i$:

$r_1 = 5.0$ cm, $\hat{r}_1 = (-0.8,\ 0.6,\ 0)$

$r_2 = 3.0$ cm, $\hat{r}_2 = (0,\ 1.0,\ 0)$

$r_3 = 5.0$ cm, $\hat{r}_3 = (0.8,\ 0.6,\ 0)$

Step 3 — Coulomb's law, $\vec{E} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r^2}\hat{r}$:

$\vec{E}_1 = (-5.75 \times 10^6,\ 4.32 \times 10^6,\ 0)$ N/C

$\vec{E}_2 = (0,\ 2.00 \times 10^7,\ 0)$ N/C

$\vec{E}_3 = \left(5.75 \times 10^6,\ 4.32 \times 10^6,\ 0\right)$ N/C

Check. P_1 and P_3 are the same distance from q. What's the same about $\vec{E}_1$ and $\vec{E}_3$, and what's different? Why?

Same magnitude (7.19×10^6 N/C) and the same $+y$ component (4.32×10^6 N/C), since P_1 and P_3 are the same height above q. They differ only in the sign of the x-component — $\vec{E}_1$ points up and to the *left*, $\vec{E}_3$ up and to the *right*. That's exactly mirror symmetry: P_1 and P_3 are mirror images of each other across the vertical line through q.

2. Biot–Savart: the same charge, now moving

Same charge, same three points — but now q moves with velocity $\vec{v} = (12, 0, 0)$ m/s. The Biot–Savart law for a single point charge is

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\,\vec{v} \times \hat{r}}{r^2}, \qquad \frac{\mu_0}{4\pi} = 1 \times 10^{-7}\ \text{T·m/A}.$$

Reuse your $\hat{r}_1, \hat{r}_2, \hat{r}_3$ from §1 — the geometry hasn't changed, only the source. Compute $\vec{v} \times \hat{r}_i$ first, then $\vec{B}_i$:

$\vec{v} \times \hat{r}_1 = \left(0,\ 0,\ 7.2\right)$ m/s

$\vec{v} \times \hat{r}_2 = \left(0,\ 0,\ 12\right)$ m/s

$\vec{v} \times \hat{r}_3 = \left(0,\ 0,\ 7.2\right)$ m/s

$\vec{B}_1 = \left(0,\ 0,\ 5.76 \times 10^{-10}\right)$ T

$\vec{B}_2 = \left(0,\ 0,\ 2.67 \times 10^{-9}\right)$ T

$\vec{B}_3 = \left(0,\ 0,\ 5.76 \times 10^{-10}\right)$ T

Check. Is $\vec{B}_1$ the mirror image of $\vec{B}_3$, the way $\vec{E}_1$ and $\vec{E}_3$ were? What's actually true about $\vec{B}_1$ and $\vec{B}_3$?

No — $\vec{B}_1$ and $\vec{B}_3$ come out *identical*, not mirror images. $\vec{v}$ points along $\hat{x}$, so $\vec{v}\times\hat{r}$ only ever depends on the y-component of $\hat{r}$ (the x-component of $\hat{r}$ drops out of that cross product entirely). $\hat{r}_1$ and $\hat{r}_3$ have the *same* y-component (0.6), so they give the same $\vec{B}$ even though their x-components have opposite signs. $\vec{B}$ simply doesn't see the left/right distinction that $\vec{E}$ did.

3. Compare $\vec{E}$ and $\vec{B}$

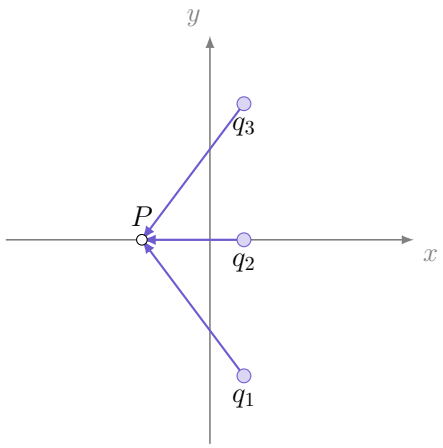
- Which field is bigger here, $\vec{E}$ or $\vec{B}$ — by roughly how many orders of magnitude? E is bigger by about 10^{16}. That's not a coincidence: $\vec{B} = (\vec{v} \times \vec{E})/c^2$, and $v/c^2 \approx 12/(9 \times 10^{16}) \approx 1.3 \times 10^{-16}$ — exactly the ratio you'd compute from B_2/E_2 above.
- $\vec{E}_1, \vec{E}_2, \vec{E}_3$ point in the same directions as $\hat{r}_1, \hat{r}_2, \hat{r}_3$ — radially away from q, since q is positive. directions (how do they relate to $\hat{r}$?).
- $\vec{B}_1, \vec{B}_2, \vec{B}_3$ all point straight out of the page, $+\hat{z}$ — because $\vec{v}$ ($\hat{x}$) and every $\hat{r}$ here (in the xy-plane) both lie in the xy-plane, and the cross product of two in-plane vectors is always perpendicular to that plane. (what direction, and why does every one of them end up that way)?

Check. $\vec{v}$ points along $+\hat{x}$ and every field point here has $z = 0$. Could $\vec{B}$ ever have come out with a nonzero x- or y-component at these points? Why or why not?

No. With $\vec{v} = (v, 0, 0)$ and $\hat{r} = (r_x, r_y, 0)$ confined to the xy-plane, $\vec{v} \times \hat{r} = (0, 0, v r_y)$ — the x- and y-components of the cross product are identically zero no matter what r_x and r_y are. $\vec{B}$ *has* to point along $\pm\hat{z}$ for any field point in this plane; only its sign and size can change.

4. Three moving charges in a line

Three charges, each $q = +1.0\ \mu\text{C}$, sit at $(1, -4, 0)$, $(1, 0, 0)$, $(1, 4, 0)$ cm, each moving with $\vec{v} = (0, 10, 0)$ m/s. Find the total $\vec{B}$ at field point $P = (-2, 0, 0)$ cm — off to the side, not on the line of charges.



Sketch the three separation vectors from each charge to P.

$\vec{r}_1 = (-3,\ 4,\ 0)$ $\quad r_1 = 5.0$ cm $\quad \hat{r}_1 = (-0.6,\ 0.8,\ 0)$

$\vec{r}_2 = (-3,\ 0,\ 0)$ $\quad r_2 = 3.0$ cm $\quad \hat{r}_2 = (-1.0,\ 0,\ 0)$

$\vec{r}_3 = (-3,\ -4,\ 0)$ $\quad r_3 = 5.0$ cm $\quad \hat{r}_3 = (-0.6,\ -0.8,\ 0)$

Each charge's contribution $\vec{B}_i = \dfrac{\mu_0}{4\pi}\dfrac{q\,\vec{v} \times \hat{r}_i}{r_i^2}$:

$\vec{B}_1 = (0,\ 0,\ 2.4 \times 10^{-10})$ T $\qquad \vec{B}_2 = (0,\ 0,\ 1.11 \times 10^{-9})$ T

$\vec{B}_3 = (0,\ 0,\ 2.4 \times 10^{-10})$ T

$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 = (0,\ 0,\ 1.59 \times 10^{-9})$ **T**

Check. Did the three contributions partly cancel, or did they add? Compare to what you'd expect from three point *charges'* $\vec{E}$ fields at an off-axis point — is $\vec{B}$ behaving the same way?

They added, all three pointing $+\hat{z}$. $\vec{v}$ here is $\hat{y}$, so $\vec{v} \times \hat{r} = (0, 0, -v\,r_x)$ — it depends only on $\hat{r}$'s x-component. All three charges sit to the *right* of P, so all three $\hat{r}_i$ have the same-sign x-component, and their contributions add instead of partly canceling. Three point *charges'* $\vec{E}$ fields at this same off-axis P would *not* all add this cleanly — E_1 and E_3 would have oppositely-signed y-components that partly cancel, the same way E_1 and E_3 differed back in §1. $\vec{B}$ is behaving differently here because the cross product throws away a different component than Coulomb's law keeps.

This is exactly why a straight current-carrying wire's field just builds up as you add more current elements, instead of partly canceling, as you saw in the sim.

5. From one moving charge to a current

A current is just a huge number of moving charges, confined to a wire. Take a short piece of wire of length $d\ell$, carrying current I: in time dt, the charge that passes through it is dq, moving at the drift velocity $\vec{v}$ along the wire. The replacement rule is

$$q\,\vec{v} \;\longrightarrow\; I\,d\vec{\ell}$$

so the Biot–Savart law for one point charge, $\vec{B} = \dfrac{\mu_0}{4\pi}\dfrac{q\,\vec{v}\times\hat{r}}{r^2}$, becomes the law for one current element:

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{\ell}\times\hat{r}}{r^2}$$

Check. Everything you did in §1–§4 above with separation vectors and unit vectors — does any of it change once $q\vec{v}$ becomes $I\,d\vec{\ell}$? What's actually new?

None of the vector mechanics changes — you still find a separation vector, divide by its length to get $\hat{r}$, cross it with a direction vector, and divide by r^2. What's new is bookkeeping, not mechanics: instead of one discrete charge's $q\vec{v}$, a real wire has charge distributed continuously along its length, so the *sum* over point charges becomes an *integral* over the wire — same ingredients, added up continuously instead of one at a time. That's exactly the sum you built by hand, element by element, in the simulator.